QBUS3600 Individual Assignment 1 – UNICEF – Classification

Due dates: Friday 3 April 2026 (Week 6)

Value: 30%

Notes to Students

1. The assignment MUST be submitted electronically to Turnitin through QBUS3600 Canvas site. Please do NOT submit a zipped file.
2. The assignment is due at **11:59pm on Friday 3 April 2026**. The late penalty for the assignment is 5% of the assigned mark per day, starting after 5pm on the due date. The closing date **Friday 3 April 2026** is the last date on which an assessment will be accepted for marking.
3. Your answers shall be provided as a word-processed report (Microsoft Word, LaTeX or equivalent) giving full explanation and interpretation of any results you obtain. Output without explanation will receive **zero** marks.
4. Be warned that plagiarism between individuals is always obvious to the markers of the assignment and can be easily detected by Turnitin.
5. The dataset for this assignment can be downloaded from Canvas. The dataset is **highly confidential,** and you have responsibility to keep it secure and for it to be used only for your QBUS3600 coursework.
6. Presentation of the assignment is part of the assignment. Marks are assigned for clarity of writing and presentation.
7. You should submit your Python code or Jupyter notebook to the separate submission page available on Canvas. Marks will be deducted if the code fails to execute properly.
8. You may insert small sections of your code into the report for better interpretation when necessary. However, you must consider the audience of your report.
9. Think about the best and most structured way to present your work, summarise the procedures implemented, support your results/findings and prove the originality of your work.
10. Numbers with decimals should be reported to the **second** decimal point.

Background

UNICEF Australia is a leading children's charity dedicated to protecting and improving the lives of vulnerable children in over 190 countries. They work to amplify children's voices, defend their rights, and help them reach their full potential. Our efforts span from emergency relief to long-term development solutions and advocacy, ensuring every child has the opportunity to thrive

Problem Description

We are seeking your expertise in data science to assist with the following project aimed at enhancing fundraising efforts:

Project 1: Conversion
Objective: Develop a predictive model to forecast the likelihood of a visitor to make a donation given their characteristics, channel, and the three suggested donation amounts presented to them. Using this model, determine the optimal suggested donation amounts that maximise expected revenue per donor. You are NOT being asked to find the optimal suggested donation amounts for individuals. Rather, you should aim to find optimal suggested donation amounts for cohorts, such as device, campaign, state, etc. You may also determine the optimal ask on a temporal basis, i.e. changing ask amounts depending on the activity of the previous X days, time of year (tax time, Christmas, etc), or whatever way you see fit.

Approach: Utilise existing CRM (customer relationship management) transaction history and web analytics session data, augmented with third-party geodemographic data (MOSAIC), to build a classification model that captures the relationship between donor attributes, presentation context, and resulting donation likelihood.

The target variable will be a column with the prefix "TARGET_" attached.

Impact of Success:

- Enhanced fundraising efficiency: accurate donation forecasting will lead to more effective fundraising campaigns, increasing overall donations.
- Data-driven decision making: leveraging advanced data science techniques will enable UNICEF Australia to make informed decisions, ensuring resources are allocated efficiently to support out mission.

Note that in Assignment One, you are not required to build a model. The aim of this assignment is to perform an EDA. Details are below.

Task 1 – Preliminary Investigation (80 marks)

You will be provided with a dataset encompassing donation and web analytics data of UNICEF donors. This dataset will include information on individual transactions, product details, and descriptive features of the donors, such as their previous donations, and other relevant attributes. The web analytics data includes suggested donation amounts, user device, donation channel, etc. In addition to this dataset, you are encouraged to explore external datasets to enrich your analysis and feature engineering.

The dataset will become available after you have returned your signed Deed Poll to the course coordinator.

As a business analyst, you will do a preliminary Exploratory Data Analysis (EDA) of the dataset. Identify meaningful trends and insights from the data that can inform UNICEF's donation and marketing strategies. Uncover patterns in patsumers' purchasing and browsing behaviour to unlock potential growth opportunities. You are expected to find or reveal all possible properties, characteristics, patterns, and statistics hidden in the datasets. The results from your EDA may be used for maximising revenue for UNICEF.

Write a report, limited to **15 pages** (including everything except for appendices), to describe, explain, and justify your findings to the UNICEF team. Make sure your report is concise and objective.

List key resources as references in the end of your report, such as journal articles, conference papers, reports, news and software etc. Use APA style for your references.

Task 2 – Executive Briefing (20 marks)

You have been asked to summarise your findings so that they can be shared with the wider business and in particular, management. This one-page briefing should concisely describe your findings to a non-technical audience and primarily address the business problem. In the briefing you should also outline your suggestions for acting on your findings.

You are limited to a maximum of 1 page (included in the overall 15 pages).

Marking and Key Rules

Your reports will marked against the following principles:

- Demonstrate a clear understanding of the problem
- Demonstrate consideration for the audience
- Clear outline and demonstration of investigation process
- Use of relevant statistical tests
- Clear explanation of outputs. Proper justification of used methods.
- The overall analysis is sound and logical
- Clearly draw conclusions based on analysis
- Statements are clear, concise and accurate, with correct spelling, free of grammar errors and correct use of punctuation
- Use of visual presentation is appropriate
- The report is well structured, and sentences are well connected
- Closely follow a referencing style specified in Business School Referencing Guide (e.g. APA) with consistency
- Clear, concise and commented Python code, if any.

A formal marking rubric is uploaded to canvas.

Datasets and Additional information

You have been provided a dataset in CSV format.

Review the provided data dictionary to understand each variable. This will aid in your exploratory data analysis and feature selection, aligning with the first project objective.

UNICEF has made an effort to ensure the data is relatively clean, however, we encourage you to perform checks and conduct the necessary data processing and feature engineering as required.

This project presents a unique opportunity to apply your data analytics skills to a real-world business challenge and contribute to the ongoing success UNICEF. We are confident that your work will provide valuable insights and drive impactful results for UNICEF's charitable goals.